

RESEARCH PROFILE

Seeking a long-term undergraduate research internship or research-assistant opportunity where I can develop rigorous research habits through implementation, reproduction, experimentation, and technical writing.

Software Engineering & Developer Tools

Design, requirements, modeling, testing, and code quality.

AI-Assisted Software Engineering

LLMs and agents for coding, testing, debugging, and workflows.

Algorithms & Reliable Systems

Retrieval, system modeling, performance, reliability, and security.

EDUCATION

East China Normal University
B.Eng. in Software Engineering, expected 2029

2025 - 2029

GPA 3.81 / 4.0	MAJOR RANK 35 / 253 - TOP 14%	SCHOLARSHIP RANK 17 / 253 - TOP 7%	CET-4 589
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SELECTED PROJECTS

ScaffoldMind
Independent Developer / Independent Project

May - Jun 2026

Designed and implemented a local-first workspace for structured computer-science learning, from product flows and interface design through backend APIs and persistence.

- Built guided chat, practice, review, history, learner-profile, and local-settings workflows.
- Implemented React and TypeScript interfaces, Express APIs, SSE streaming, and backend LLM integration.
- Kept API credentials server-side and learning records local.

[React](#) / [TypeScript](#) / [Node.js](#) / [Express](#) / [SSE](#) / [LLM API](#)

Mini-VDB: Dynamic Vector Index & Exact Top-K Retrieval
Independent Developer / Data Structures and Algorithm Practice / Supervisor: Wang Liping

Jun 2026

Implemented the indexing and retrieval core of a small vector database and documented design trade-offs for exact similarity search under dynamic updates.

- Supported insertion, deletion, exact Top-K, and threshold retrieval with Euclidean distance and inner product.
- Implemented open-address hashing, heap-based Top-K selection, KD-tree pruning, and AVX2 sequential scanning.
- Developed eight progressive versions spanning sorting, priority queues, BSTs, and AVL trees.

[C++](#) / [KD-tree](#) / [Top-K Retrieval](#) / [Heap](#) / [AVX2](#)

Intelligent Elevator Collaborative Modeling & Simulation
Control Modeling Contributor / Five-person course project / Supervisor: Liu Jing

Jul 2026

Independently completed the continuous-dynamics and control portion of a collaborative SysML and Simulink elevator model.

- Modeled car, counterweight, load, gravity, friction, braking, and acceleration-to-position dynamics.
- Implemented PID feedback and load compensation with gravity, acceleration, and friction feed-forward terms.
- Validated final position error below 0.01 m for no-load, half-load, and full-load simulations.

[MATLAB](#) / [Simulink](#) / [SysML](#) / [PID Control](#) / [Dynamic Modeling](#)

ADDITIONAL PROJECT

Maple Database & Core API Subtask Owner / Four-person project Owned the persistence and core-data-service subtask for a local mind-map notebook with editing and version-management features. - Designed node, connection, and metadata schemas with Spring Boot entity mappings. - Implemented APIs for node CRUD, connection management, and layout-data storage. Java 17 / Spring Boot / JPA / H2 / SQLite / REST API / Git	Jan - Mar 2026
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COMPETITION & DESIGN WORK

Intelligent Water Quality Monitoring & Early-Warning System Team Lead / Four-person technical proposal / Oct - Nov 2025 Coordinated the submission and contributed to the proposed sensing, acquisition, communication, data-fusion, warning, and energy architecture. Deliverables were a technical proposal, functional diagrams, and device concept design.	Zhiyi Technical & Functional Proposal Design / Ten-person project / Jun 2026 Authored the technical and functional sections of an education-technology proposal, translating educational goals into software functions, interaction flows, technical architecture, and application scenarios.
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HONORS & AWARDS

Undergraduate Excellence Scholarship First Class, East China Normal University	2025-2026
Advanced Robotics and Simulation Technology Competition National Third Prize	Nov 2025
Challenge Cup Entrepreneurship Plan Competition, ECNU Round Silver Award	Jun 2026
ECNU Coder Freshman Programming Challenge Third Prize	2025
Boyuan IT Club Owner-Pro Third Prize	Mar 2026

TECHNICAL SKILLS

LANGUAGES	C++, C, Python, Java, TypeScript
AI & RETRIEVAL	LLM application development, prompt engineering, vector search, PyTorch
SYSTEMS & TOOLS	Linux, Git/GitHub, unit testing, REST APIs
MODELING	MATLAB, Simulink, SysML, dynamic-system modeling

SELECTED COURSEWORK

Data Structures and Algorithms - 4.0/4.0 - 98/100	Linear Algebra - 4.0/4.0
Computer Systems - 4.0/4.0 - 90/100	Mathematics for Information Security - 4.0/4.0
Object-Oriented Programming (C++) - 4.0/4.0	Mathematical Thinking of AI - 4.0/4.0
Discrete Mathematics - 4.0/4.0	Data Structures and Algorithm Practice - 4.0/4.0